AUTOMATA THEORY
SAMPLE PRACTICE QUESTIONS

Q1. Convert the following CFG to CNF

$S \rightarrow ABaC \mid Ba$ $\quad B \rightarrow bB \mid b$

$A \rightarrow aC \mid CC$ $\quad C \rightarrow d \mid \varepsilon$

Mention all the steps and the corresponding rules used at appropriate positions

Q2. Construct a Deterministic PDA for the followings:

i) Accepting set of all strings over {0, 1} which do not contain 1001.

ii) $L = \{w1cw2: \text{ such that } |w_1| > |w_2| \text{ and } w \varepsilon \{a,b\}^*\}$.

Validate your answers

Q3. Construct a CFG accepting all strings over {0, 1} which contain the substring 1101.

Q4. Identify the language of the given grammar:

$S \rightarrow AbB$

$A \rightarrow aA \mid a$

$B \rightarrow bB \mid b$

Q5. Simplify the following CFG.

$S \rightarrow ABC \mid aABa$ $\quad B \rightarrow bB \mid CC$ $\quad K \rightarrow ABt$

$A \rightarrow aC \mid CB$ $\quad C \rightarrow d \mid \varepsilon$

Mention all the steps and the corresponding rules used at appropriate positions

Q6. Construct a PDA for the followings:

i) A Language accepting set of all strings over {a, b} which contain substring abab.

ii) CFG given in Q5.

Validate your answers